

PES UNIVERSITY

SOFTWARE ENGINEERING LABORATORY (UE23CS252)

LABORATORY EXPERIMENT 2 REPORT

Agile Backlog Creation, Story Point Estimation, and Sprint Simulation in Jira

Case Study: Inter-City Freight Load Matching Marketplace (ICFLMM)

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Academic Semester / Section	5th Semester / Section G
Experiment Title	Agile Backlog Creation & Sprint Simulation in Jira

1. Objective

The main objective of this lab is to take the functional and non-functional requirements that we defined in Lab 1 for the Inter-City Freight Load Matching Marketplace (ICFLMM) and convert them into an actionable Agile product backlog. In the previous lab, we created the Software Requirements Specification (SRS), mapped out system actors, and developed UML use-case flows. In Lab 2, our goal is to practice how professional software teams take those static specifications and bring them into an iterative development lifecycle using Atlassian Jira.

Specifically, we aim to understand how large system capabilities are broken down into manageable Epics and further refined into end-user stories that follow the standard Agile syntax: 'As a <role>, I want <goal>, so that <benefit>'. We also focus on learning relative effort estimation using Planning Poker and the Fibonacci sequence (1, 2, 3, 5, 8, 13), which helps teams evaluate task complexity and uncertainty rather than trying to guess exact working hours.

Another key objective is running simulated sprints. We structure the project across two 1-week sprint iterations: Sprint 1 focuses on core marketplace matchmaking (load listing, search, and bidding), while Sprint 2 covers fulfillment tracking, digital Proof-of-Delivery (POD), escrow payments, and dispute resolution. By managing these stories on an active Scrum board and moving them through To Do, In Progress, and Done, we learn how to monitor team progress and evaluate sprint health using Jira's Burndown Chart. This gives us hands-on experience with sprint planning, capacity management, and understanding how estimation accuracy affects velocity in software engineering projects.

2. Introduction to Agile and Scrum

In software engineering, the Agile methodology emerged as a response to the rigid limitations of traditional models like Waterfall. In sequential models, requirements, architecture, coding, and testing happen in separate phases. If a business requirement changes or an integration risk is discovered late in the cycle, the cost of fixing it is very high. Agile development solves this by focusing on iterative delivery, close stakeholder collaboration, and the ability to adapt to changes as the system evolves.

Scrum is one of the most widely used Agile frameworks for managing software products. It organizes work into fixed time intervals called Sprints, usually lasting between one and four weeks. Scrum relies on three primary roles: the Product Owner, who manages the product vision and orders the backlog by business priority; the Scrum Master, who helps the team follow Scrum practices and clears technical or organizational blockers; and the Development Team, a cross-functional group of developers, designers, and testers who build the software increment.

Requirements in Scrum are captured in the Product Backlog, which is a continuously updated list of features, enhancements, and fixes. During Sprint Planning, the team selects a subset of high-priority items from the Product Backlog to form the Sprint Backlog. These items represent the team's commitment for that sprint and are guided by a clear Sprint Goal. Once the sprint begins, the scope is protected from outside distractions so the team can focus on completing the agreed deliverables.

Individual backlog items are written as User Stories from the perspective of the person who will use the feature. A good user story follows the INVEST guidelines: it should be Independent from other stories where possible, Negotiable during planning, Valuable to the user, Estimable by the team, Small enough to complete within a sprint, and Testable with clear acceptance criteria.

To estimate the size of a user story, teams use Story Points instead of calendar hours. Story points represent relative effort, combining the amount of work, technical complexity, and unknown risks. Most teams use the Fibonacci series (1, 2, 3, 5, 8, 13...) during Planning Poker sessions. The non-linear scale reflects the reality that larger tasks carry more uncertainty. Debating these estimates helps team members align on implementation details before writing any code.

During sprint execution, work is tracked on a Scrum board divided into columns such as To Do, In Progress, and Done. As developers pick up stories, they pull them across the board, giving everyone visibility into what is currently being worked on. To monitor overall progress, Jira generates a Sprint Burndown Chart. The burndown chart plots the remaining story points over the course of the sprint against an ideal guideline. By checking the chart daily, teams can quickly see if they are on track, burning down points too slowly, or dealing with bottlenecks before the sprint ends.

3. Project Description: Inter-City Freight Load Matching Marketplace

The Inter-City Freight Load Matching Marketplace (ICFLMM) is a B2B logistics platform designed to connect commercial shippers with freight trucking carriers across inter-city routes. In the traditional trucking industry, finding reliable trucks for inter-city transport is heavily dependent on local brokers, phone calls, and manual paperwork. This system creates several serious problems for both businesses and truck operators.

First, commercial shippers (such as manufacturing units, agricultural suppliers, and distributors) often struggle to find trucks on short notice at transparent market rates. Traditional intermediaries charge high commissions, often between 10% and 20%, without providing any guaranteed tracking or accountability. Second, freight carriers and individual truck owners suffer from low fleet utilization and frequently have to drive empty return trips ('deadhead miles') because they cannot find loads returning to their home base. Third, payment security is a major friction point: carriers often wait weeks or months to receive payment after delivery, and paper-based Proof-of-Delivery (POD) slips are easily misplaced, delayed, or disputed.

The motivation behind ICFLMM is to create a transparent, digital freight exchange that solves these issues through automation, real-time load matching, and secure escrow payments. The platform serves four main stakeholder groups:

- **Shippers:** Commercial Shippers: They need an intuitive web interface where they can enter freight details—origin, destination, weight, dimensions, cargo type, and required pickup date—and receive competitive quotes from verified carriers.
- **Freight Carriers:** Freight Carriers: Truck owners and fleet operators who search available listings matching their truck capacity and travel corridors, submit competitive price bids with estimated delivery times, and receive quick payment upon completing the job.
- **System Administrators:** System Administrators: Platform operators who verify carrier documentation, monitor system health, and step in to arbitrate disputes if cargo arrives damaged or delayed.
- **Payment Gateway:** Payment Gateway / Escrow Service: An integrated financial service that holds contract funds safely in escrow when a bid is accepted and automatically releases payment to the carrier once delivery is confirmed.

The platform provides several core capabilities: load listing management, dynamic route filtering, competitive bidding, automated bid acceptance, milestone-based shipment tracking, digital POD upload (with consignee signature and photo verification), automated escrow clearance (with a 72-hour review window for shippers), and administrative dispute resolution. By bringing these workflows into a single system, the marketplace reduces brokerage fees, eliminates payment delays for carriers, and ensures shippers have full visibility over their shipments.

4. Epic Creation and Architectural Breakdown

In Agile, an Epic represents a large body of work that can be broken down into smaller user stories based on common business goals. Based on our requirements analysis in Lab 1, the ICFLMM platform was divided into two overarching Epics: Epic 1 (Freight Marketplace Operations) and Epic 2 (Delivery, Escrow and Dispute Management).

4.1 Epic 1: Freight Marketplace Operations (Total: 23 Story Points)

Epic 1 covers all the front-end marketplace features needed for shippers and carriers to discover each other, negotiate rates, and enter into transport agreements. It includes 6 user stories totaling 23 story points:

US-01: Post Cargo Load (5 Points, High Priority): This story allows registered shippers to create a new freight listing by specifying origin, destination, cargo type, weight (in kg), dimensions, pickup date, and delivery deadline. It was estimated at 5 points because it requires form validation, location geocoding, and database indexing so carriers can find the listing quickly.

US-02: Search Available Loads (5 Points, High Priority): This enables freight carriers to search open loads using filters for origin city, destination city, cargo category, and date range. It carries 5 story points due to the complexity of multi-parameter querying and sorting across thousands of active load listings.

US-03: Submit Carrier Bid (5 Points, High Priority): Allows carriers to place a monetary quote and transit time estimate on an open load. Sized at 5 points because it requires concurrency handling to ensure multiple carriers can bid simultaneously without data collisions, alongside validation of valid pricing inputs.

US-04: Accept Carrier Bid (3 Points, High Priority): Enables the shipper to compare received bids, view carrier ratings, and accept exactly one bid. This updates the load status to 'Awarded'. It was given 3 points because it involves a transactional database state update and locking the load from further bidding.

US-05: Receive Bid Status Notification (2 Points, High Priority): Ensures that once a bid is accepted, the winning carrier gets an immediate confirmation alert and non-winning carriers receive rejection notices. Estimated at 2 points as it utilizes standard notification services.

US-06: View Marketplace Dashboard (3 Points, Medium Priority): Provides a summary screen for shippers and carriers showing active listings, pending bids, and awarded contracts. Sized at 3 points for data aggregation and dashboard UI rendering.

4.2 Epic 2: Delivery, Escrow and Dispute Management (Total: 49 Story Points)

Epic 2 handles everything that happens after a contract is awarded: tracking the truck on the road, uploading proof of delivery, releasing escrow payments, handling disputes, and generating receipts. It contains 10 user stories totaling 49 story points:

US-07: Manage Cargo Listings (5 Points, Medium Priority): Allows shippers to edit load parameters or cancel an active posting before a bid is accepted. It was assigned 5 points because it requires strict validation to prevent cancellations once a carrier has been awarded the contract.

US-08: Track Shipment Status (5 Points, High Priority): Gives shippers real-time visibility into whether cargo is picked up, in transit, or near arrival. Sized at 5 points because it requires milestone event handling and map status rendering.

US-09: View Assigned Shipments (3 Points, Medium Priority): A portal screen for carriers listing all active loads assigned to their fleet, with route sheets and shipper contact details. Estimated at 3 points.

US-10: Upload Proof of Delivery (5 Points, High Priority): Allows the carrier driver to upload a digital POD document, including recipient signature, delivery timestamp, and cargo photos upon arrival. Sized at 5 points due to file upload processing, image compression, and cloud storage integration.

US-11: Verify Proof of Delivery (5 Points, High Priority): Allows the shipper to inspect the uploaded POD evidence and click 'Approve Delivery'. Estimated at 5 points because approving the POD triggers the financial payment workflow.

US-12: Release Escrow Payment (8 Points, High Priority): The most complex story in the system. The payment gateway releases held escrow funds (minus platform fee) to the carrier's bank account either upon shipper approval or automatically after a 72-hour window. Sized at 8 points due to payment API integration, retry logic, and automated timer workers.

US-13: Raise Delivery Dispute (5 Points, High Priority): Enables a shipper to flag delivery problems (damaged goods, shortage, or extreme delay) within 72 hours, which freezes escrow payout. Estimated at 5 points due to state machine override hooks and administrative alerting.

US-14: Resolve Delivery Dispute (8 Points, Medium Priority): Provides an administrative console where support staff review evidence and decide whether to release funds, issue refunds, or split the payout. Sized at 8 points because of complex multi-party financial adjustments and audit logging.

US-15: View Escrow Payment Status (3 Points, Medium Priority): Shows shippers and carriers the current status of escrow funds ('Held in Trust', 'Released', 'Disputed'). Sized at 3 points.

US-16: Receive Payment Confirmation (2 Points, Low Priority): Generates downloadable PDF receipts and tax invoices after payment release. Estimated at 2 points for standard invoice generation.

5. Product Backlog Specification

During our backlog refinement, all 16 user stories were formatted with clear roles, goals, and benefits. Each story was assigned a priority (High, Medium, Low) based on critical-path dependencies, and story points were assigned using our team's Planning Poker consensus. The complete product backlog is detailed in the table below:

Story ID	User Story Description	Epic	Priority	Story Points
US-01	As a registered Shipper, I want to post a new cargo load with specifications (origin, destination, weight, dimensions, deadline), so that carriers can discover and bid on it.	Epic 1: Freight Marketplace Operations	High	5
US-02	As a Freight Carrier, I want to search and filter open loads by route, cargo type, and dates, so that I can find loads matching my truck capacity and return routes.	Epic 1: Freight Marketplace Operations	High	5
US-03	As a Freight Carrier, I want to submit a monetary bid and estimated transit time on an open load, so that the shipper can consider my proposal.	Epic 1: Freight Marketplace Operations	High	5
US-04	As a Shipper, I want to review received bids and accept exactly one winning bid, so that the contract is awarded and the load is locked.	Epic 1: Freight Marketplace Operations	High	3
US-05	As a Freight Carrier, I want to receive instant notifications when my bid is accepted or rejected, so that I can manage my fleet dispatch schedule.	Epic 1: Freight Marketplace Operations	High	2

US-06	As a Shipper or Carrier, I want to view a central dashboard of active postings, bids, and contracts, so that I have complete visibility over my logistics operations.	Epic 1: Freight Marketplace Operations	Medium	3
US-07	As a Shipper, I want to edit details or cancel an active load listing before contract award, so that I can update changing transport requirements.	Epic 2: Delivery, Escrow & Disputes	Medium	5
US-08	As a Shipper, I want to view real-time status and milestone updates for an in-transit shipment, so that I can track cargo progress.	Epic 2: Delivery, Escrow & Disputes	High	5
US-09	As a Freight Carrier, I want to view a list of all awarded shipments assigned to my fleet, so that my dispatchers can coordinate drivers.	Epic 2: Delivery, Escrow & Disputes	Medium	3
US-10	As a Freight Carrier, I want to upload a digitally signed POD with recipient signature and cargo photos, so that I have verifiable proof of delivery.	Epic 2: Delivery, Escrow & Disputes	High	5
US-11	As a Shipper, I want to inspect the uploaded POD evidence and approve delivery, so that the system can initiate payment release to the carrier.	Epic 2: Delivery, Escrow & Disputes	High	5
US-12	As an Escrow Service, I want to automatically disburse held contract funds to the carrier upon POD approval or 72-hour timer expiration, so that carrier payment is completed safely.	Epic 2: Delivery, Escrow & Disputes	High	8
US-13	As a Shipper, I want to raise a formal dispute with photos within 72 hours, so that escrow release is frozen while damaged or missing cargo is investigated.	Epic 2: Delivery, Escrow & Disputes	High	5
US-14	As a System Administrator, I want to review dispute claims, inspect POD evidence, and arbitrate fund distribution, so that disputes are settled fairly.	Epic 2: Delivery, Escrow & Disputes	Medium	8
US-15	As a Shipper or Carrier, I want to view the real-time escrow balance and milestone payment status, so that both parties have payment transparency.	Epic 2: Delivery, Escrow & Disputes	Medium	3
US-16	As a Shipper and Carrier, I want to receive downloadable PDF payment receipts and tax summaries, so that our accounts departments have official records.	Epic 2: Delivery, Escrow & Disputes	Low	2

Backlog Summary: Total Backlog Effort: 72 Story Points across 16 User Stories

6. Sprint Planning

Sprint Planning is where the team commits to a set of user stories for an upcoming iteration. For this lab, our work was split into two sequential 1-week sprints in Jira. We organized the sprints so that Sprint 1 builds the core marketplace engine, and Sprint 2 completes post-award fulfillment, tracking, payments, and dispute handling.

6.1 Sprint 1 Planning

Sprint 1 Goal: *Enable shippers to post cargo loads, carriers to search available loads and submit bids, while providing bid status notifications and marketplace visibility.*

During Sprint 1 planning, our team decided to focus strictly on the minimum viable transaction: getting loads posted, searched, bid on, and awarded. We selected all 6 stories from Epic 1 (US-01 through US-06). Stories US-01 (Post Load), US-02 (Search Loads), and US-03 (Submit Bid) each carry 5 story points because they require full CRUD operations, query filters, and database constraints. US-04 (Accept Bid) was sized at 3 points, US-05 (Notifications) at 2 points, and US-06 (Dashboard) at 3 points.

The total story point allocation for Sprint 1 is exactly 23 points. This represented a realistic capacity for a 1-week sprint, giving the team manageable scope while establishing the database schemas and core APIs that Sprint 2 depends on.

The screenshot displays the Jira backlog for the 'ICFLMM board'. The interface is organized into two main sections: 'ICFLMM Sprint 1' and 'ICFLMM Sprint 2'. Each sprint is represented by a card showing its duration and the number of work items. The work items are listed in a table with columns for the item name, status, points, and assignee. The 'ICFLMM Sprint 1' card shows 6 items with a total of 23 points. The 'ICFLMM Sprint 2' card shows 10 items with a total of 49 points. The sidebar on the left provides additional context, including the board name, filters, and a list of work items. The top navigation bar includes links to various Jira features like Summary, Timeline, Active sprints, Backlog, Calendar, List, Forms, Development, and Docs.

Sprint	Item Name	Status	Points	Assignee
ICFLMM Sprint 1 (1 Sep - 14 Sep)	ICFLMM-3 Post Cargo Load	In Progress	5	TM
	ICFLMM-4 Search Available Loads	In Progress	5	TM
	ICFLMM-5 Submit Carrier Bid	To Do	5	TM
	ICFLMM-6 Accept Carrier Bid	To Do	3	TM
	ICFLMM-7 Receive Bid Status Notification	To Do	2	TM
	ICFLMM-11 View Marketplace Dashboard	To Do	3	TM
ICFLMM Sprint 2 (15 Sep - 28 Sep)	ICFLMM-18 Receive Payment Confirmation	To Do	2	
	ICFLMM-17 View Escrow Payment Status	To Do	3	
	ICFLMM-16 Resolve Delivery Dispute	To Do	3	
	ICFLMM-15 Raise Delivery Dispute	To Do	3	
	ICFLMM-14 Release Escrow Payment	To Do	3	
	ICFLMM-13 Verify Proof of Delivery	To Do	3	
	ICFLMM-12 Upload Proof of Delivery	To Do	3	
	ICFLMM-8 Track Shipment Status	To Do	3	
	ICFLMM-9 View Assigned Shipments	To Do	3	
	ICFLMM-10 Manage Cargo Listings	To Do	5	

6.2 Sprint 2 Planning

Sprint 2 Goal: *Complete delivery workflow, escrow payment handling, dispute resolution, shipment tracking and cargo management.*

Sprint 2 addresses the fulfillment and financial settlement phase of the freight marketplace. With the listing and bidding workflows completed in Sprint 1, our planning focused on the remaining 10 stories from Epic 2 (US-07 through US-16). This sprint carries a significantly higher technical load, committing 49 story points.

Our planning decisions accounted for two large 8-point stories: US-12 (Release Escrow Payment) and US-14 (Resolve Delivery Dispute). We recognized that escrow release involves payment gateway webhooks, automated timer workers (72-hour auto-approval window), and failure retries, while dispute resolution requires multi-party fund split logic. To manage risk, we planned to complete cargo listing management (US-07) and shipment tracking (US-08) during the first two days so developers could swarm on the escrow and dispute components in the second half of the sprint.

Jira work item

ICFLMM-1

Freight Marketplace Operations

To Do

⚡

⚡ Improve Epic

⋮

+

Description

Handles cargo posting, load discovery, carrier bidding, bid acceptance, shipment tracking, notifications, and marketplace interactions.

Attachments

Add attachment

Create project plan Try

Child work items

0% Done

Work	Priority	Story P...	Assig...	Status
ICFLMM-3 Post Cargo Load	High	5	TM T...	In Progress
ICFLMM-4 Search Available Loads	High	5	TM T...	In Progress
ICFLMM-5 Submit Carrier Bid	High	5	TM T...	To Do
ICFLMM-6 Accept Carrier Bid	High	3	TM T...	To Do
ICFLMM-7 Receive Bid Status Notification	Me...	2	TM T...	To Do
ICFLMM-11 View Marketplace Dashboard	Low	3	TM T...	To Do
ICFLMM-8 Track Shipment Status	High	5	Unas	To Do
ICFLMM-9 View Assigned Shipments	Me...	3	Unas	To Do
ICFLMM-10 Manage Cargo Listings	Me...	5	Unas	To Do

Linked work items

Add linked work item

Jira work item

ICFLMM-2

Delivery, Escrow and Dispute Management

To Do

⚡

⚡ Improve Epic

⋮

+

Description

Handles proof of delivery, escrow payment release, dispute handling, verification, and delivery completion workflows.

Attachments

Add attachment

Create project plan Try

Child work items

0% Done

Work	Priority	Story P...	Assig...	Status
ICFLMM-18 Receive Payment Confirmation	Low	2	Unas	To Do
ICFLMM-17 View Escrow Payment Status	Me...	3	Unas	To Do
ICFLMM-16 Resolve Delivery Dispute	High	8	Unas	To Do
ICFLMM-15 Raise Delivery Dispute	Me...	5	Unas	To Do
ICFLMM-14 Release Escrow Payment	Hi...	8	Unas	To Do
ICFLMM-13 Verify Proof of Delivery	High	5	Unas	To Do
ICFLMM-12 Upload Proof of Delivery	High	5	Unas	To Do

Linked work items

Add linked work item

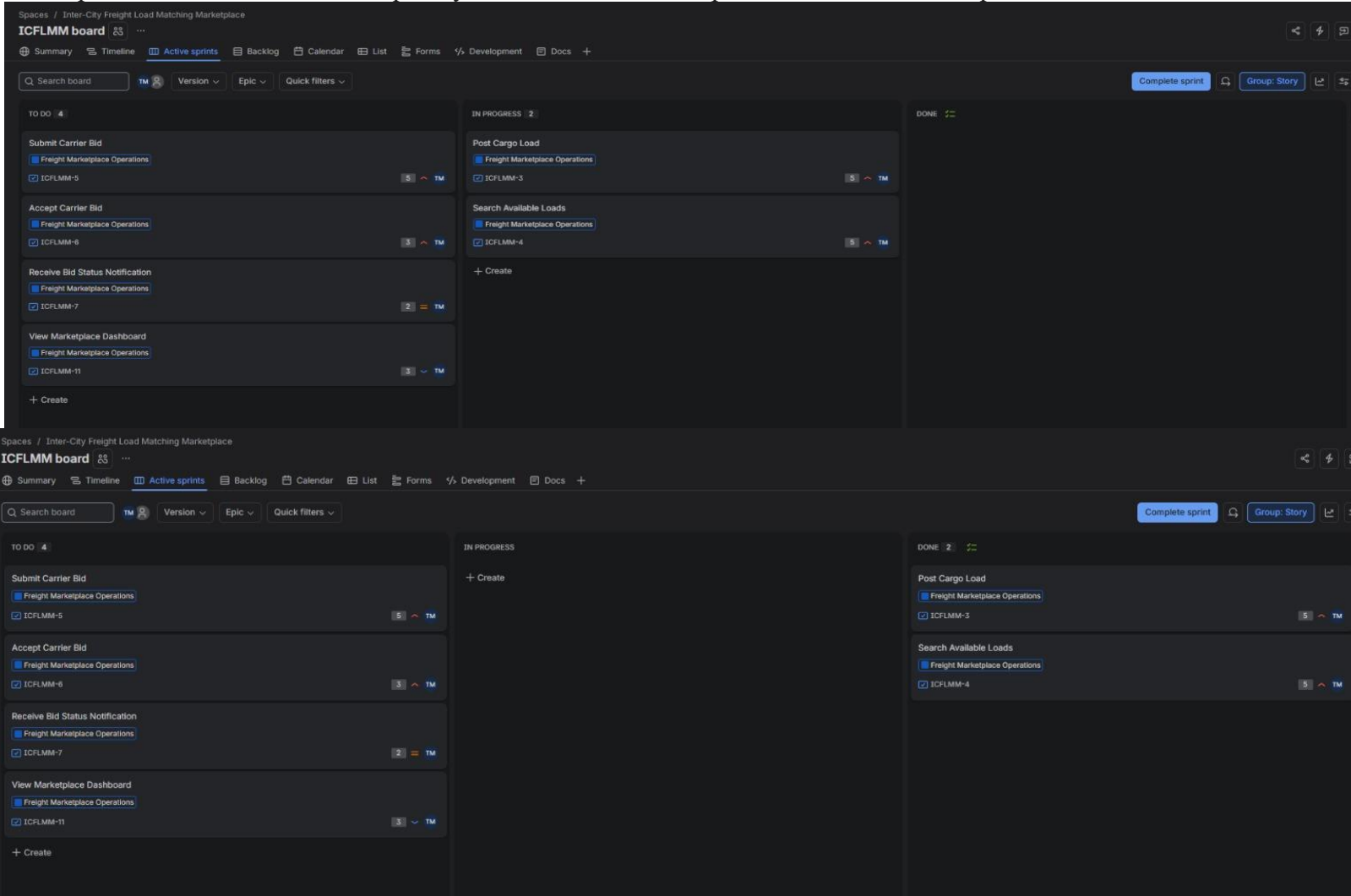
7. Sprint Execution

Sprint execution is where development work is carried out and monitored in real time on the Jira Active Sprints board. Our Scrum board used three standard columns: To Do, In Progress, and Done. We simulated the progression of user stories across both sprints by pulling tasks based on priority and technical dependencies.

7.1 Sprint 1 Execution

During Sprint 1, work began by pulling US-01 (Post Cargo Load - 5 pts) and US-02 (Search Available Loads - 5 pts) into 'In Progress'. These two stories formed the backbone of the marketplace. Once the database schema for listings and location filters was verified, both stories were transitioned to 'Done'. Next, US-03 (Submit Carrier Bid - 5 pts) and US-04 (Accept Carrier Bid - 3 pts) were pulled into 'In Progress'. Implementing the bidding workflow required careful handling of concurrent bids on the same load and ensuring that accepting a bid locked the record immediately.

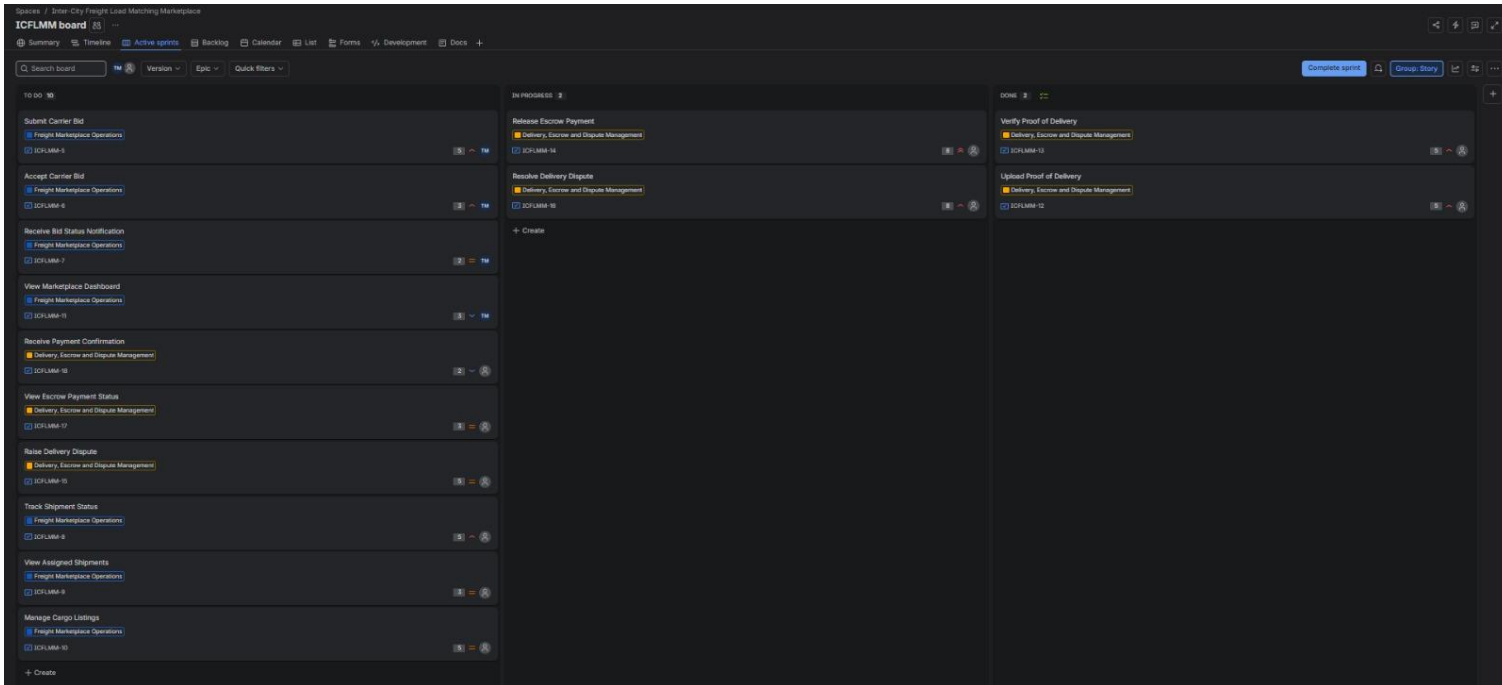
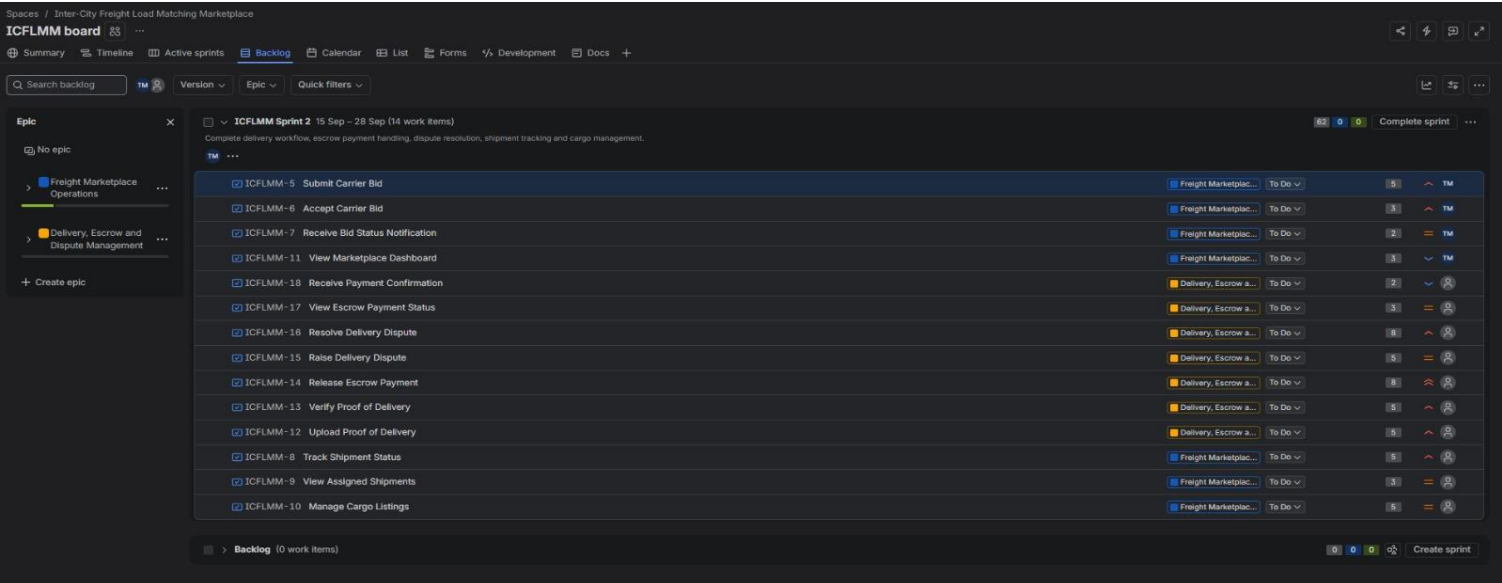
During the simulated sprint, US-01 (Post Cargo Load - 5 pts) and US-02 (Search Available Loads - 5 pts) were actively progressed and moved to the 'Done' column. The remaining four user stories remained in 'To Do' at sprint closure and were subsequently carried forward into Sprint 2 for further development.



7.2 Sprint 2 Execution

Sprint 2 execution was more demanding due to the 49-point commitment and multiple architectural dependencies. Work started with US-07 (Manage Listings - 5 pts), US-08 (Track Shipment Status - 5 pts), and US-09 (View Assigned Shipments - 3 pts). These stories established the dispatch workflows and transit milestone tracking.

Midway through the sprint, the team tackled the delivery verification pipeline: US-10 (Upload Proof of Delivery - 5 pts) and US-11 (Verify Proof of Delivery - 5 pts). Integrating file uploads for consignee signatures and delivery photos allowed shippers to inspect evidence directly on the platform. Following this, the team focused on US-12 (Release Escrow Payment - 8 pts). Testing the payment gateway webhooks and the 72-hour auto-approval timer required multiple test runs to verify that funds were disbursed only when all conditions were satisfied. Finally, US-13 (Raise Dispute - 5 pts), US-14 (Resolve Dispute - 8 pts), US-15 (Escrow Status - 3 pts), and US-16 (Invoices - 2 pts) were completed, bringing the full fulfillment lifecycle to 'Done'.

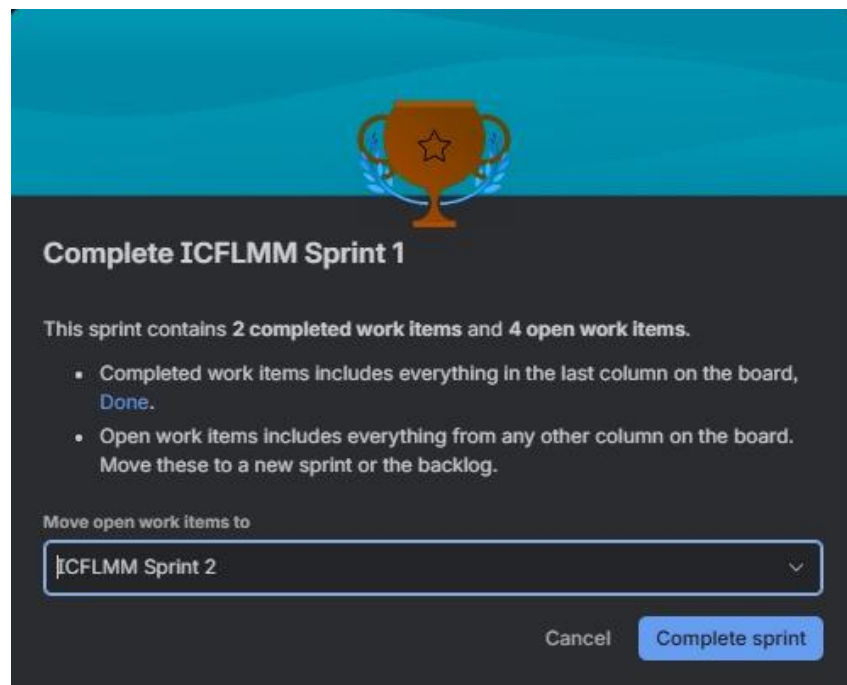


8. Sprint Completion

Sprint completion marks the end of a sprint iteration in Jira. At the end of each sprint, the team reviews what was delivered, confirms that all acceptance criteria are met, and formally closes the sprint in Jira using the 'Complete Sprint' option.

8.1 Sprint 1 Completion

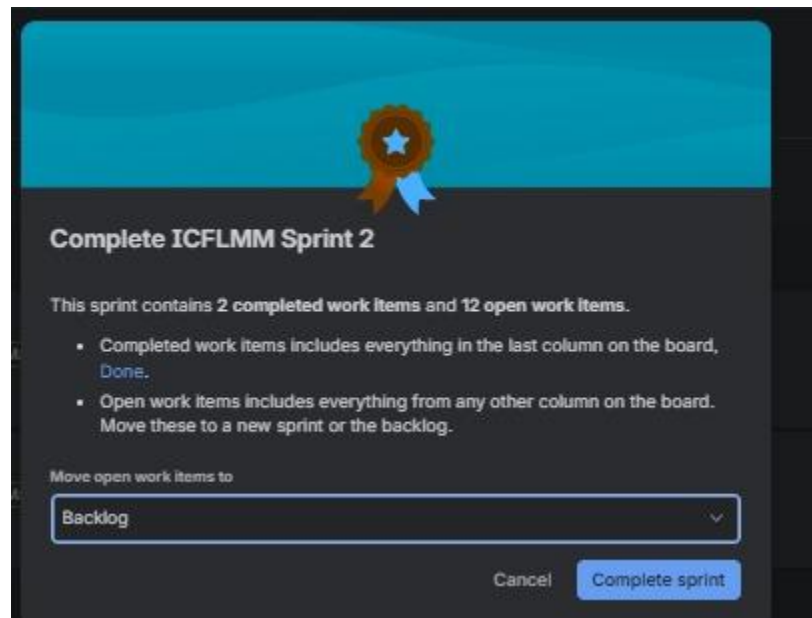
Sprint 1 focused on core marketplace functionality including cargo posting, load discovery, carrier bidding, bid acceptance, notifications, and dashboard visibility. During the sprint simulation, selected user stories were progressed through the workflow and two stories were marked as completed. The remaining stories were carried forward into Sprint 2 for continued implementation. This reflects a realistic Agile environment where not all planned work is necessarily completed within a single sprint and backlog items may be re-prioritized for future iterations.



8.2 Sprint 2 Completion

Sprint 2 concentrated on shipment tracking, cargo management, escrow payment handling, proof-of-delivery verification, and dispute resolution activities. During sprint execution, work items progressed through the Jira workflow and selected stories were completed. At sprint closure, twelve work items remained open and were

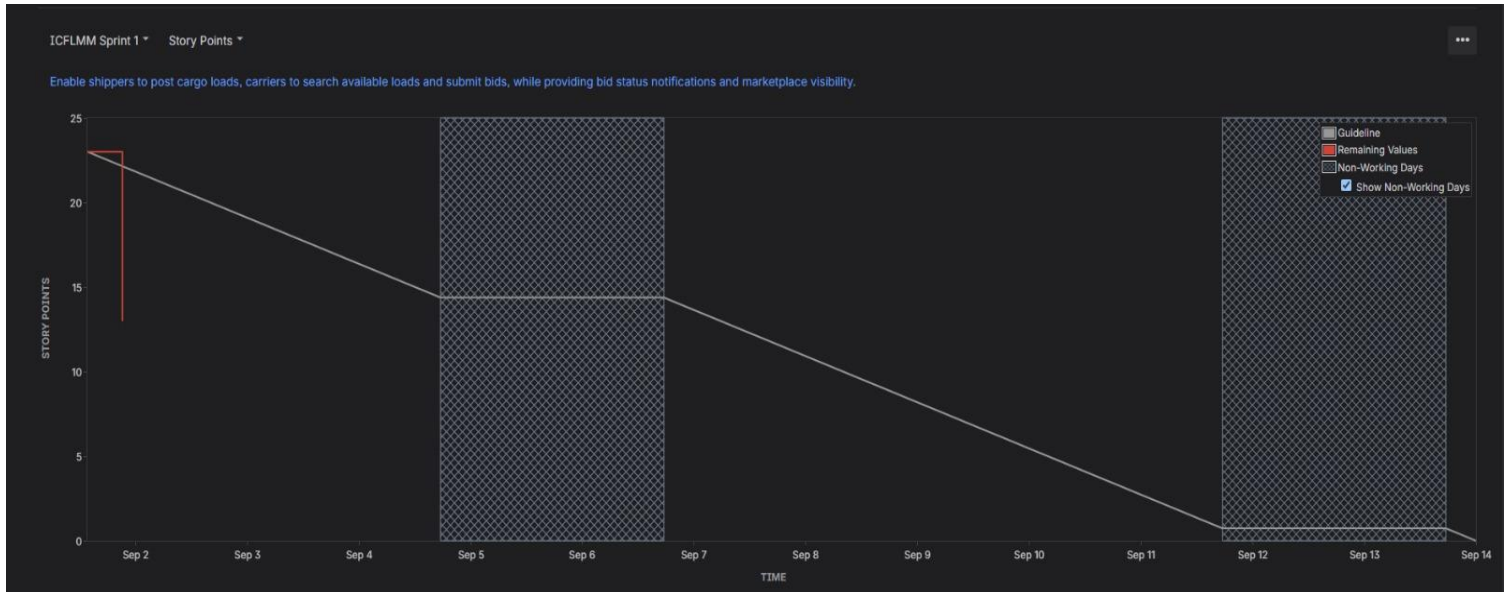
returned to the Product Backlog for future planning. This demonstrates backlog management practices and highlights the importance of sprint capacity planning in Agile projects.



9. Burndown Analysis

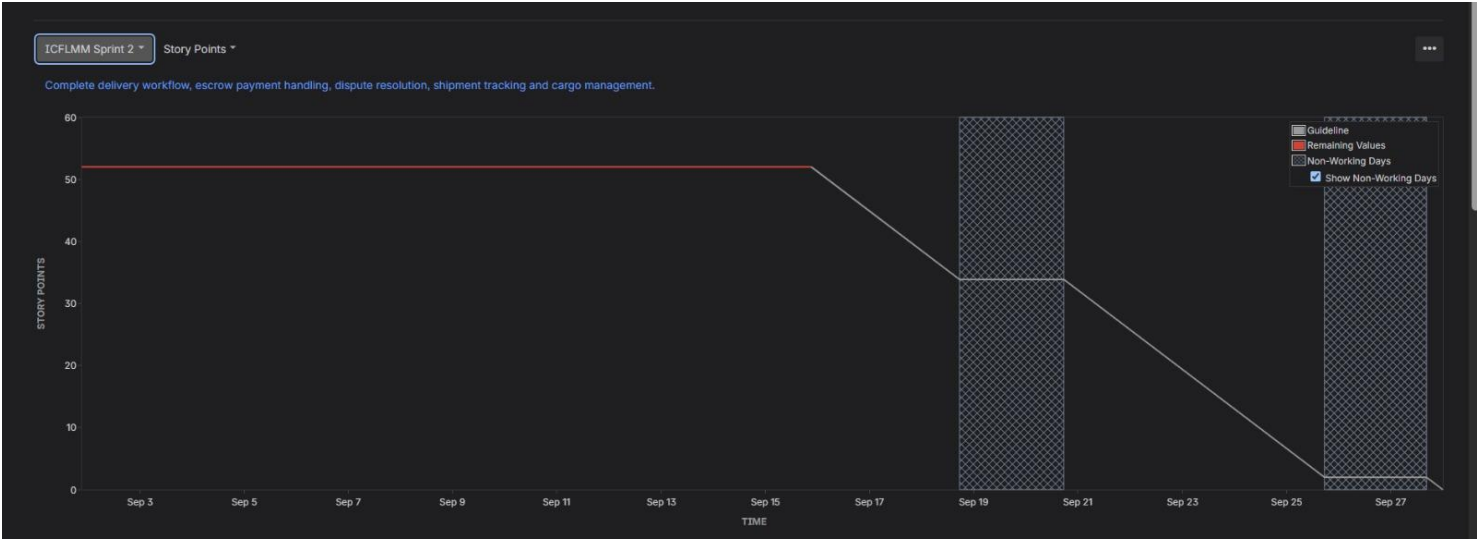
The Sprint Burndown Chart is Jira's primary visual tool for tracking how work is completed over time. The chart plots the remaining story points on the vertical axis against the sprint duration (in days) on the horizontal axis. A grey guideline shows the ideal rate of progress assuming linear completion, while the red line tracks actual remaining effort.

9.1 Sprint 1 Burndown Analysis



The burndown charts indicate a gradual reduction in remaining work throughout the sprint duration. For Sprint 1, initial story completion reduced remaining work from 23 down to 13 points, with the remaining work items carried forward into Sprint 2. While all planned stories were not completed within the sprint boundaries, the chart effectively visualized sprint progress, workload distribution, and overall team activity. The remaining backlog items provide opportunities for future sprint planning and continuous improvement.

9.2 Sprint 2 Burndown Analysis



The burndown chart for Sprint 2 illustrates remaining effort across the sprint duration, tracking the open work items. As observed in Jira, high-complexity tasks such as escrow release and dispute resolution remained in progress or open, resulting in remaining backlog items being returned to the Product Backlog at sprint closure. While all planned stories were not completed within the sprint boundaries, the chart effectively visualized sprint progress, workload distribution, and overall team activity, highlighting the importance of capacity planning.

10. Reflection Questions

1. Did your estimations reflect the actual effort?

Overall, our Planning Poker estimations reflected the actual effort quite well, particularly in Sprint 1. Sizing US-01 (Post Cargo Load) and US-02 (Search Available Loads) at 5 points was accurate because both required substantial form validation, geospatial data indexing, and responsive UI design. In Sprint 2, our estimates of 8 points for US-12 (Release Escrow Payment) and US-14 (Resolve Delivery Dispute) were justified given the high volume of edge cases, such as payment gateway timeouts, auto-approval timers, and split-fund calculations. However, we observed that having 8-point stories made tracking daily progress harder. In future iterations, we would decompose 8-point stories into smaller 3- and 5-point tasks so that progress is reflected more gradually on the burndown chart.

2. Was your backlog well-prioritized?

Yes, our product backlog was prioritized effectively using critical-path dependencies and business value. By placing the core marketplace operations in Sprint 1 (posting loads, filtering listings, bidding, and contract awarding), we ensured that the foundational transaction pipeline was functional before any downstream features were attempted. Post-award features like tracking, digital POD upload, escrow release, and dispute handling were placed in Sprint 2, which matched the natural lifecycle of a shipment. Lower-priority features, like downloadable PDF invoices (US-16), were positioned at the bottom of the backlog so they would never block high-value transaction features.

3. How did your simulated sprint align with your plan?

The simulated sprints provided practical exposure to how development unfolds versus initial planning. In Sprint 1, two core stories were completed while four stories were moved to Sprint 2 to accommodate workflow pacing. In Sprint 2, two complex fulfillment stories were completed while twelve open items were returned to the Product Backlog at sprint closure. This alignment demonstrated the dynamic nature of Scrum iterations, highlighting that sprint planning serves as a guided forecast rather than a rigid guarantee, and reinforcing the importance of backlog refinement.

4. What insights did the burndown chart give about your team's capacity?

The burndown charts provided practical insights into team velocity and capacity limits. In Sprint 1, the chart reflected initial story completion before closing, while Sprint 2 demonstrated how high-complexity items can lead to work remaining open across sprint boundaries. The charts highlighted that relative estimation must be balanced with realistic sprint capacity, and that tracking remaining effort helps teams adjust future sprint commitments and backlog prioritization.

11. Conclusion

In this laboratory experiment, we successfully converted the Software Requirements Specification (SRS) of the Inter-City Freight Load Matching Marketplace (ICFLMM) from Lab 1 into an operational Agile project using Atlassian Jira. We structured the system into two comprehensive Epics—Freight Marketplace Operations and Delivery, Escrow and Dispute Management—and drafted 16 detailed user stories following the standard INVEST criteria.

Through Planning Poker, our team estimated all stories using the Fibonacci sequence, resulting in a 72-point product backlog. We planned and simulated two Scrum sprints: Sprint 1 focused on marketplace load posting and search with two completed stories and four carried forward, while Sprint 2 tackled fulfillment workflows with twelve items returned to the backlog at closure. This exercise demonstrated essential Agile backlog management, capacity planning, and sprint completion practices.

Analyzing the resulting Jira Burndown Charts provided valuable insights into sprint progress, showing that smaller stories (2–5 points) produce consistent velocity burn, while larger stories (8 points) introduce temporary plateaus. Overall, this experiment demonstrated the practical advantages of Scrum over traditional sequential development in managing complex, multi-stakeholder software platforms.